Linking somatic mutation in cancer to the electronic properties of DNA

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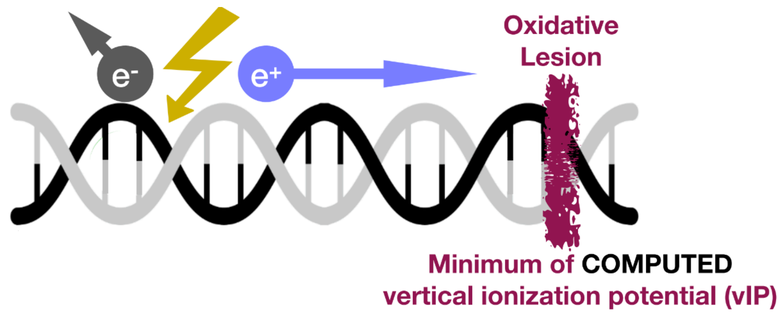
Abstract



Oxidative stress generates reactive species that damage DNA, creating lesions that can become mutations if not repaired, some of which can contribute to cancer development. Not all DNA regions are equally vulnerable: their susceptibility depends in part on the DNA sequence and its intrinsic electronic properties. Here, we show[1,2] that mutability is linked to the **vertical Ionization Potential (vIP)** of short DNA motifs, which measures how easily electrons can be removed from the nucleobase stack, with low-vIP sequences more easily oxidized. Oxidative damage hotspots and mutation-prone DNA regions can thus be understood as locations with low vIP. This perspective links DNA physics to mutagenesis, providing a mechanistic explanation for why certain genomic regions accumulate oxidative damage and mutations.

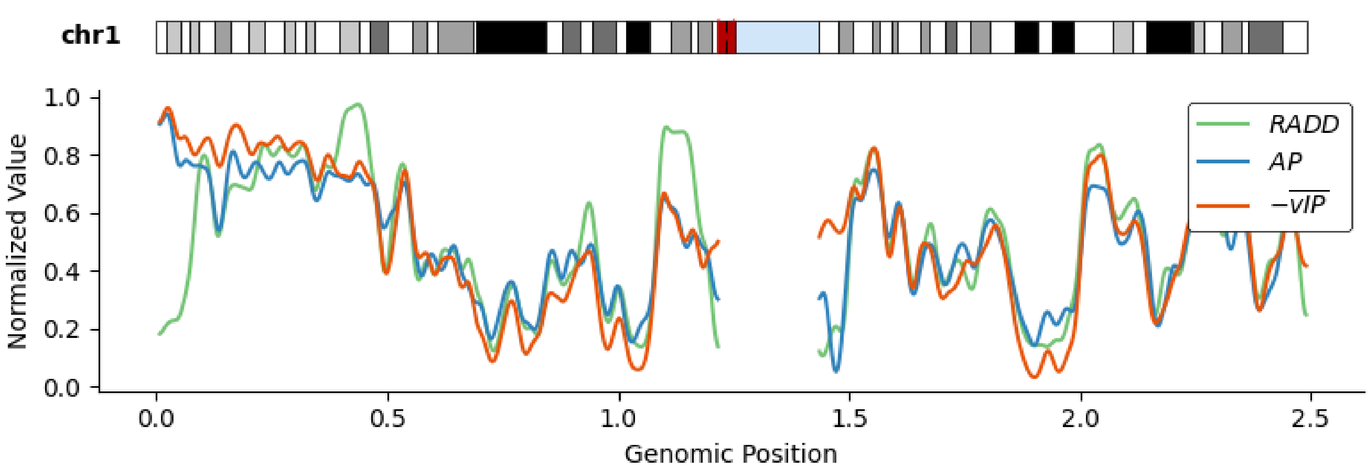
Preprint

Genome-scale oxidation damage data and vIP

The vIP of all possible DNA motifs have been computed using the **vIPer**[3] algorithm, which combines quantum chemistry calculations and mathematical modelling

We analyzed data generated by two high-throughput techniques designed to profile oxidative DNA damage genome-wide: **repair-assisted damage detectection (RADD)** sequencing performed on DNA oxidized with potassium bromate[4], and **apurinic-site (AP)** sequencing following X-ray exposure[5].

We observe a high variation in mutability across the entire genome. The highest levels of experimentally measured damage occur in regions with lowest calculated **vIP** values.



Normalized RADD, AP and (-$\overline{\mathrm{vIP}}$) scores along chromosome 1.

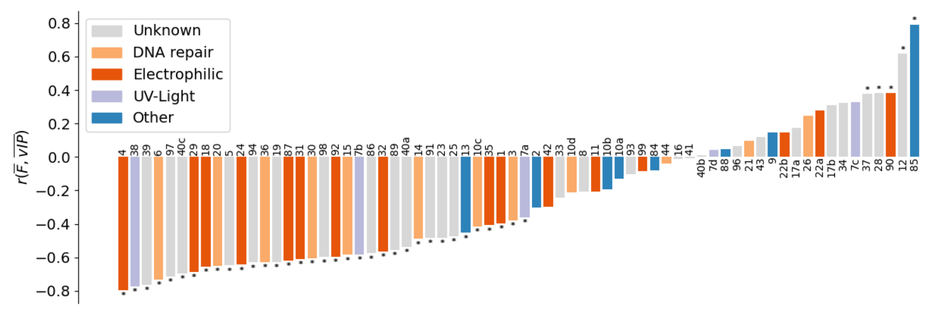
The agreement between (-**vIP**) values and experimental mutability scores indicates that DNA electronic properties play a key role in the mutability process, **supporting a causal connection between these two non-local processes.**

DNA oxidative damage and mutational signature

We considered the cancer mutation signatures reported in the COSMIC database[6]. Each signature corresponds to a different mutation aetiology. Some are directly related to DNA oxidation process, others not.

For each signature, we computed the Pearson correlatio between the logarithm of the normalized mutation frequency (F) of the nucleobase triplets and their calculated **vIP** values (using vIPer[3]).

Most signatures resulting from oxidative damage show an F-vIP anticorrelation, *e.g.,* signatures due to electophilic substances and some UV-related or DNA repair-related signatures.



F-vIP Pearson correlations for COSMIC mutational signatures[6]

Example: Signatures **SBS4**, **SBS92**, **SBS55**, **SBS29** show strong anticorrelations; they are annotated as caused by tobacco carcinogens known to induce oxidative DNA damage.

Comparison between somatic and germline mutations

We computed the correlation between the vIP of mutated triplet motifs `NXN` (`X` mutated nucleobase, `N` any base) and the logarithm of the normalized mutation frequency ($\overline{\mathrm{F}}$) in three datasets:

S_{SC}: **Somatic Cancer** mutations; S_{SN}: **Somatic Normal** mutations; S_G: **Germline** mutations

In coding regions: $\overline{\mathrm{vIP}}$ - $\overline{\mathrm{F}}$ anticorrelations are strongest for silent mutations, followed by missense and nonsense mutations.
In non-coding regions: $\overline{\mathrm{vIP}}$ - $\overline{\mathrm{F}}$ are strongest in UTR, but also significant in introns, splice regions and intergenic regions.

- Negative selection seems to shape the mutation spectrum: Highly deleterious mutations tend to be removed from cell populations, which likely explains why severe mutation types tend to show weaker anticorrelations than generaly more benign ones.
- The vIP-F anticorrelations are very similar for somatic and germline mutations, as well as in cancer and normal tissues. This suggests that **these mutations arise through similar mechanisms, in which oxidative damage plays a central role.**

Triplets - NXN	S_{SC}	S_{SN}	S_G	Total
silent	−0.86	−0.87	−0.86	−0.87
missense	−0.77	−0.76	−0.72	−0.77
nonsense	−0.69	−0.63	−0.62	−0.69
intronic	−0.67	−0.71	−0.74	−0.69
UTR	−0.75	−0.75	−0.75	−0.76
splice region	−0.60	−0.68	−0.63	−0.63
intergenic	−0.72	−0.72	−0.80	−0.72

F-vIP Pearson correlations

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